Onesimus Director — Comparison Report

Date: 20260329-150428

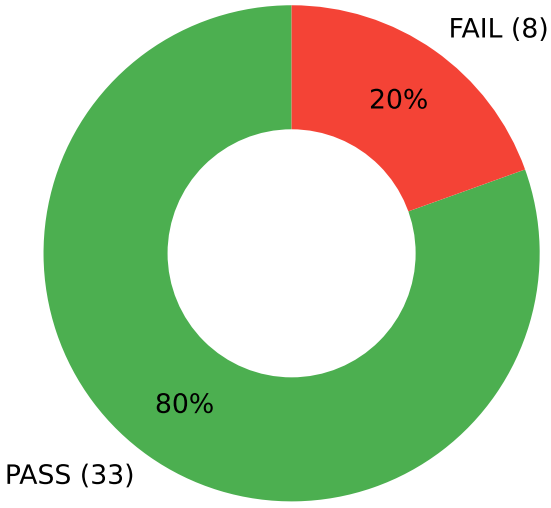
```
Host:     WS-JOERG.intranet.bernau.family
OS:       Linux-6.12.74+deb13+1-amd64-x86_64-with-glibc2.41
Kernel:   6.12.74+deb13+1-amd64
CPU:      AMD Ryzen 9 9950X 16-Core Processor
RAM:      60.4 GB
Python:   3.13.5
```

```
C++ Director:
  Port: 9101
  unknown

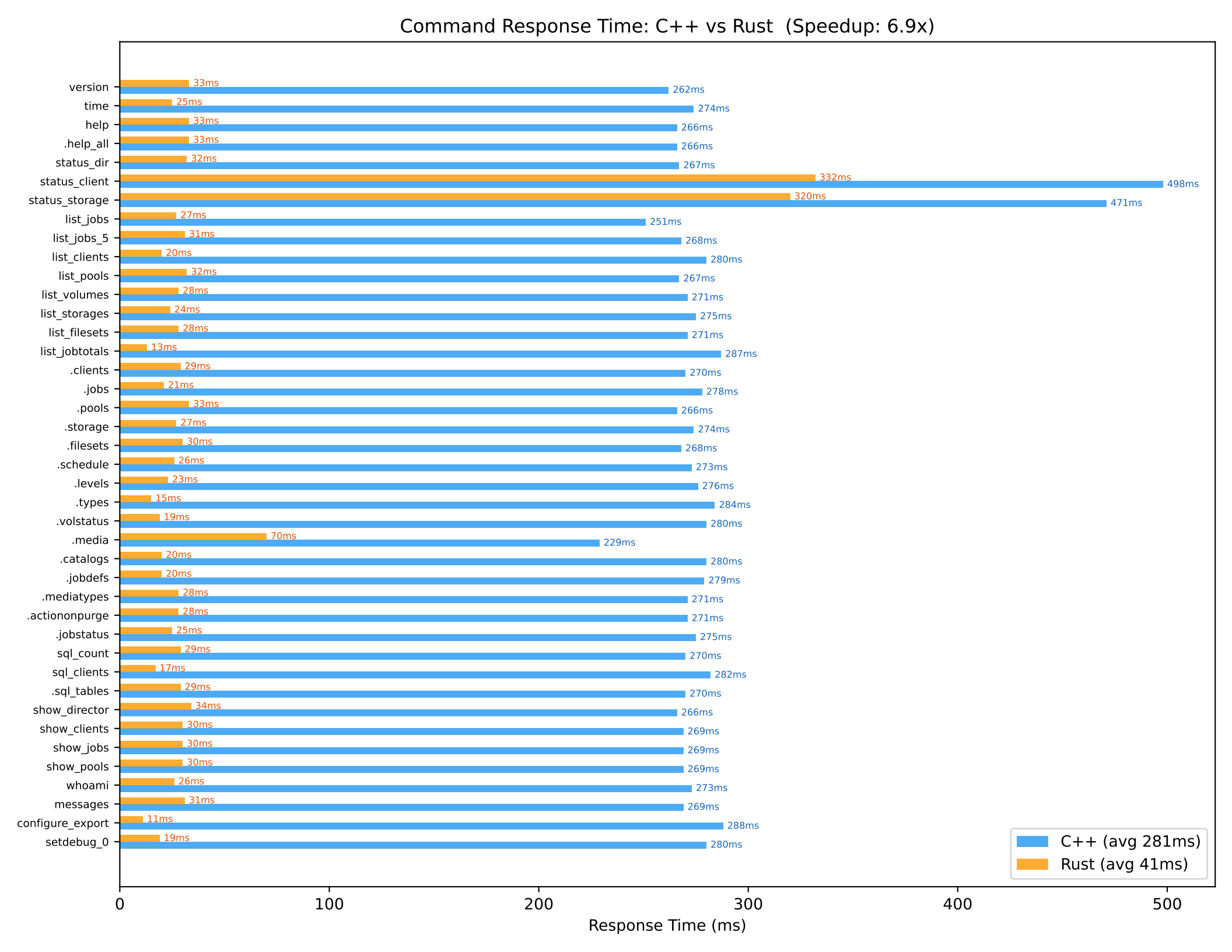
Rust Director:
  Port: 29101
  onesimusd 0.25.0
```

```
  Total: 41    Passed: 33 (80%)    Failed: 8    Skipped: 0
Avg Response:  C++ = 281 ms    Rust = 41 ms    Speedup: 6.9x
```

Method: bconsole pipes command → stdout captured → ANSI stripped → timestamps/versions/IDs normalized → sorted line comparison




```
Failure Analysis:
  help: Version string difference — different binaries, expecte
  .help_all: Version string difference — different binaries, expecte
  list_filesets: Output format difference — Rust shows more/fewer detail
  .jobs: Output content mismatch — check diff for details.
  .filesets: Output content mismatch — check diff for details.
  show_director: Port difference (C++ on 9101, Rust on 29101) — expected
  show_jobs: Output format difference — Rust shows more/fewer detail
  configure_export: Port difference (C++ on 9101, Rust on 29101) — expected
```


Command Response Time: C++ vs Rust (Speedup: 6.9x)
version
time
help
.help_all
status_dir
status_client
status_storage
list_jobs
list_jobs_5
list_clients
list_pools
list_volumes
list_storages
list_filesets
list_jobtotals
.clients
.jobs
.pools
.storage
.filesets
.schedule
.levels
.types
.volstatus
.media
.catalogs
.jobdefs
.mediatypes
.actiononpurge
.jobstatus
sql_count
sql_clients
.sql_tables
show_director
show_clients
show_jobs
show_pools
whoami
messages
configure_export
setdebug_0
33ms
262ms
25ms
274ms
33ms
266ms
33ms
266ms
32ms
267ms
332ms
498ms
320ms
471ms
27ms
251ms
31ms
268ms
20ms
280ms
32ms
267ms
28ms
271ms
24ms
275ms
28ms
271ms
13ms
287ms
29ms
270ms
21ms
278ms
33ms
266ms
27ms
274ms
30ms
268ms
26ms
273ms
23ms
276ms
15ms
284ms
19ms
280ms
70ms
229ms
20ms
280ms
20ms
279ms
28ms
271ms
28ms
271ms
25ms
275ms
29ms
270ms
17ms
282ms
29ms
270ms
34ms
266ms
30ms
269ms
30ms
269ms
30ms
269ms
26ms
273ms
31ms
269ms
11ms
288ms
19ms
280ms
C++ (avg 281ms)
Rust (avg 41ms)
0
100
200
300
400
500
Response Time (ms)

Test Results Detail (Page 1)

Test	Match	C++	Rust	Comment
version	yes	262ms	33ms	
time	yes	274ms	25ms	
help	structure_diff	266ms	33ms	Version string difference — different binaries, expected.
.help_all	structure_diff	266ms	33ms	Version string difference — different binaries, expected.
status_dir	yes	267ms	32ms	
status_client	yes	498ms	332ms	
status_storage	yes	471ms	320ms	
list_jobs	yes	251ms	27ms	
list_jobs_5	yes	268ms	31ms	
list_clients	yes	280ms	20ms	
list_pools	yes	267ms	32ms	
list_volumes	yes	271ms	28ms	
list_storages	yes	275ms	24ms	
list_filesets	structure_diff	271ms	28ms	Output format difference — Rust shows more/fewer detail line
list_jobtotals	yes	287ms	13ms	
.clients	yes	270ms	29ms	
.jobs	no	278ms	21ms	Output content mismatch — check diff for details.
.pools	yes	266ms	33ms	
.storage	yes	274ms	27ms	
.filesets	no	268ms	30ms	Output content mismatch — check diff for details.
.schedule	yes	273ms	26ms	
.levels	yes	276ms	23ms	
.types	yes	284ms	15ms	
.volstatus	yes	280ms	19ms	
.media	yes	229ms	70ms	
.catalogs	yes	280ms	20ms	
.jobdefs	yes	279ms	20ms	
.mediatypes	yes	271ms	28ms	
.actiononpurge	yes	271ms	28ms	
.jobstatus	yes	275ms	25ms	

Test Results Detail (Page 2)

Test	Match	C++	Rust	Comment
sql_count	yes	270ms	29ms	
sql_clients	yes	282ms	17ms	
.sql_tables	yes	270ms	29ms	
show_director	partial	266ms	34ms	Port difference (C++ on 9101, Rust on 29101) — expected when
show_clients	yes	269ms	30ms	
show_jobs	structure_diff	269ms	30ms	Output format difference — Rust shows more/fewer detail line
show_pools	yes	269ms	30ms	
whoami	yes	273ms	26ms	
messages	yes	269ms	31ms	
configure_export	structure_diff	288ms	11ms	Port difference (C++ on 9101, Rust on 29101) — expected when
setdebug_0	yes	280ms	19ms	

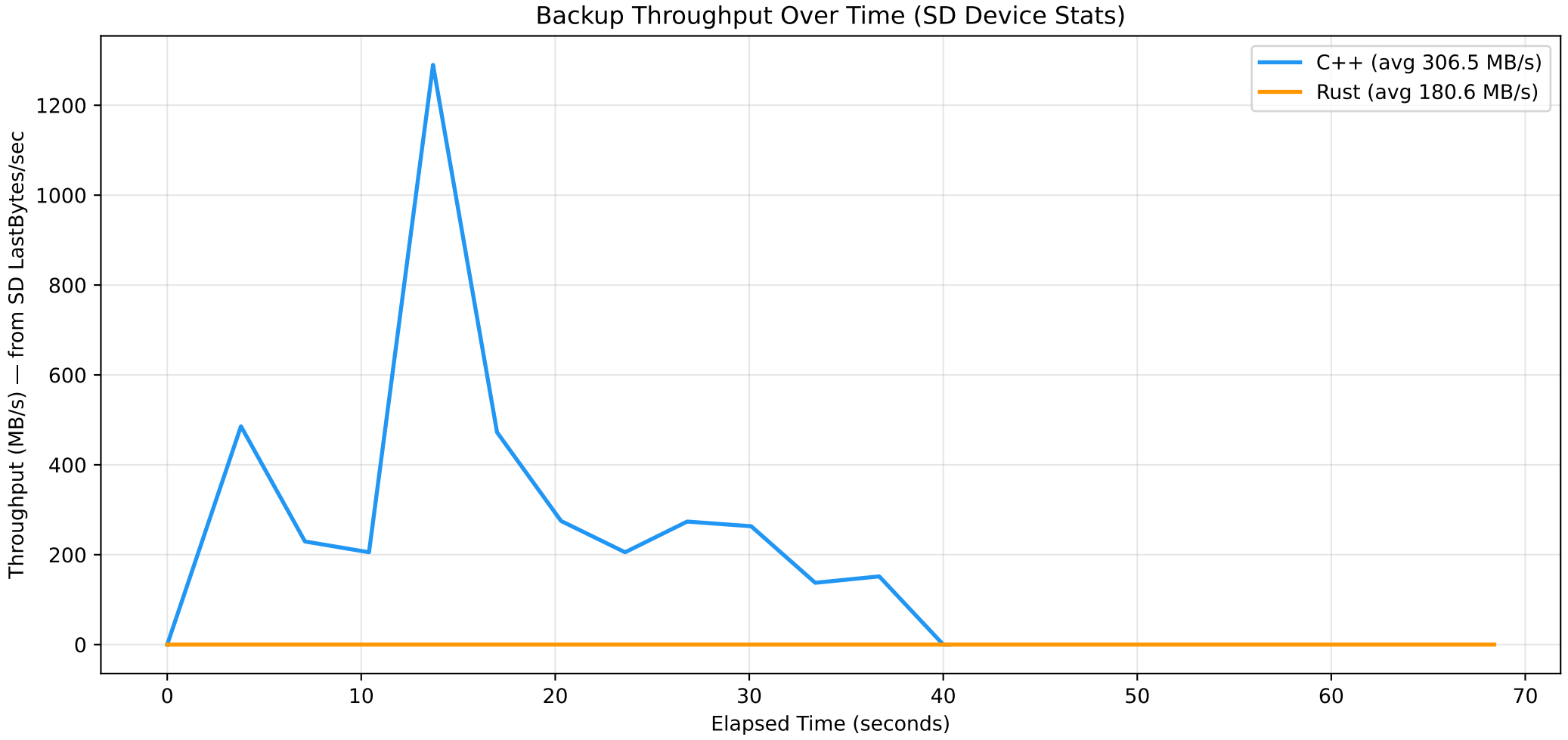
Onesimus Director — Backup Throughput Benchmark

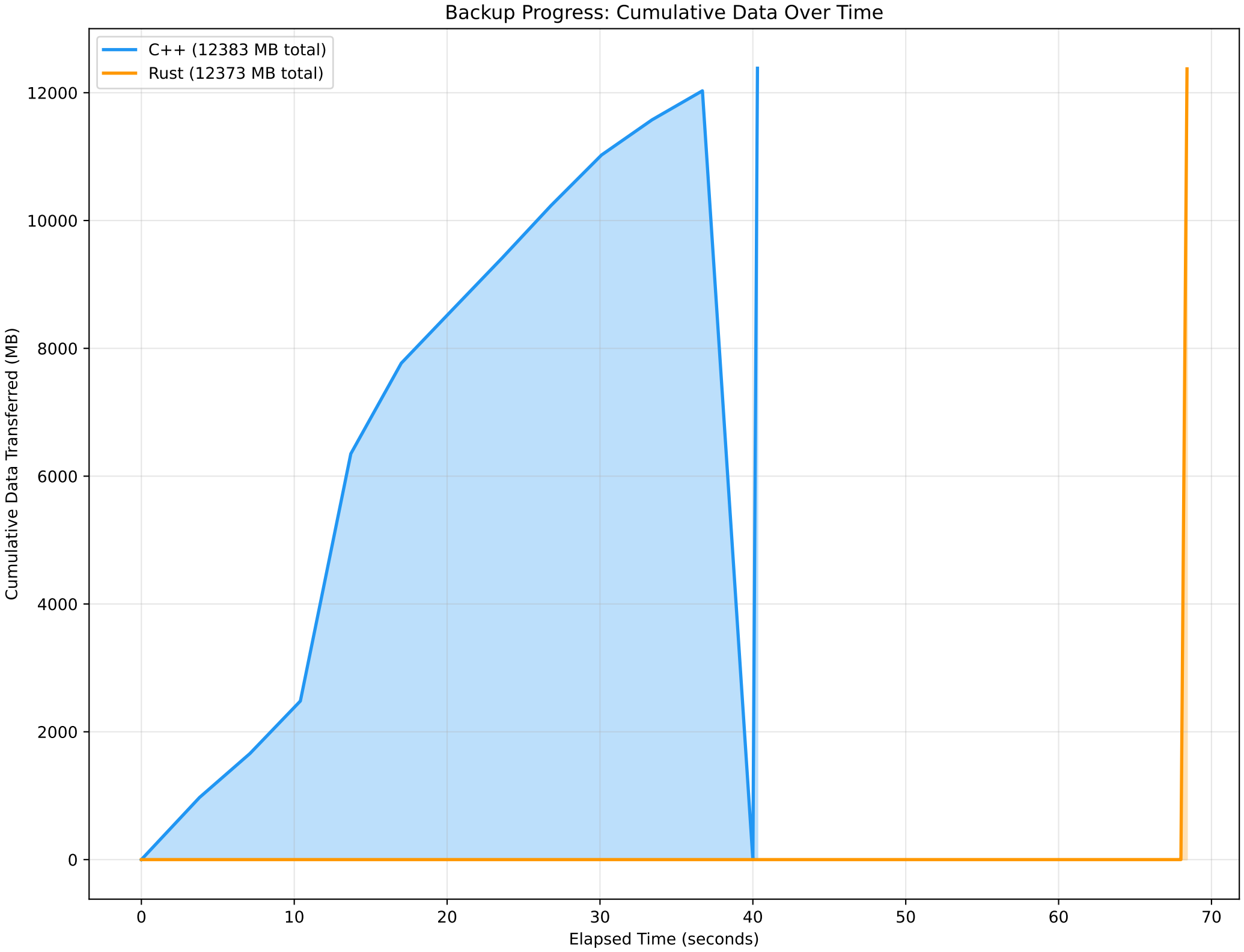
Host: WS-JOERG-MINI Platform: debian family OS: Linux-6.12.74+deb13+1-amd64-x86_64-with-glibc2.41
CPU: AMD Ryzen 9 9950X 16-Core Processor RAM: 60.4 GB Arch: x86_64

FileSet: /usr/bin, /usr/sbin, /usr/share
Files: 166,733 Directories: 15,091 Total size: 4.03 GB

Test conditions: Both Directors ran sequentially (never parallel) on the same host, same config, same catalog DB, same FD/SD. No other applications were running during the test.

	C++ Director	Rust Director	Delta
JobId			
Status	T	T	
Files	181,887	0	
Data (MB)	12,383	12,373	
Duration (s)	40.4	68.5	0.6x
Avg Throughput	306.5 MB/s	180.6 MB/s	0.6x




Backup Progress: Cumulative Data Over Time
C++ (12383 MB total)
Rust (12373 MB total)
Cumulative Data Transferred (MB)
0
2000
4000
6000
8000
10000
12000
0
10
20
30
40
50
60
70
Elapsed Time (seconds)